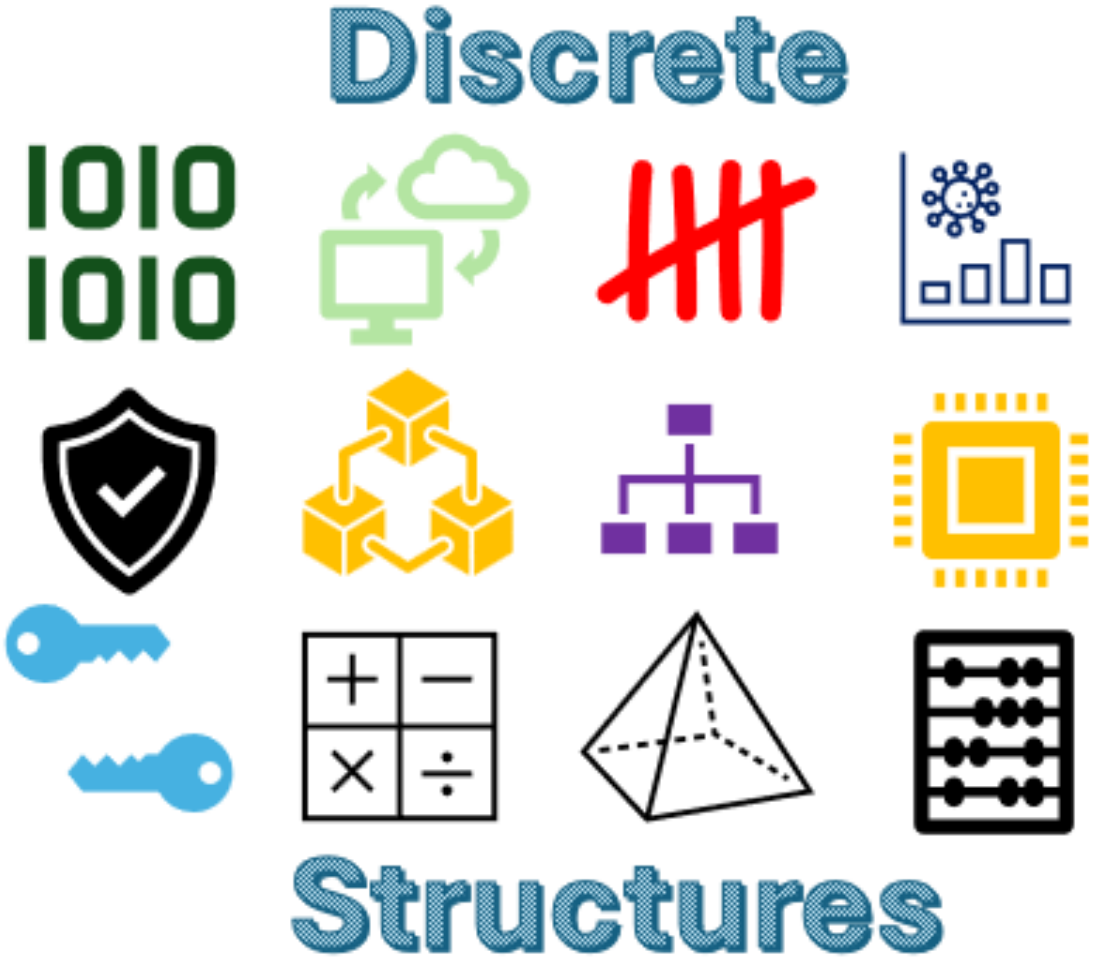
COMP 2310: Discrete Structures
Fall 2026

Chapter 3: Counting



Recurrence relations: definitions and example

- **Recurrence relation:** recursive formula that defines a sequence
- **Components of recurrence relation:**
  - **Recurrence equation:** How each term is defined using previous term
  - **Initial condition:** specifies the initial value of the sequence
  - **Example:**
    - Recurrent equation: $a_n = a_{n-1} + 3$
    - Initial condition (base case): $a_1 = 4$
    - Next numbers: 7, 10, 13,….

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- **Components of recurrence relation:**
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  - **Example:**
    - Recurrent equation for Fibonacci sequence: $f_n = f_{n-1} + f_{n-2}$,
    - Initial condition (base case): $f_1 = f_2 = 1$
    - Next numbers: 2, 3, 5, 8

Recurrence relations: finding explicit formula

- **Example:**
  - Recurrent equation: $a_n = a_{n-1} + 3$
  - Initial condition (base case): $a_1 = 4$
  - Next numbers/sequence?

$$\begin{aligned} a_n &= a_{n-1} + 3 \\ &= (a_{n-2} + 3) + 3 \\ &= ((a_{n-3} + 3) + 3) + 3 \end{aligned}$$

or

$$\begin{aligned} a_n &= a_{n-1} + 3 \\ &= a_{n-2} + 2 \cdot 3 \\ &= a_{n-3} + 3 \cdot 3 \end{aligned}$$

Eventually this process will produce

$$\begin{aligned} a_n &= a_{n-(n-1)} + (n-1) \cdot 3 \\ &= a_1 + (n-1) \cdot 3 \\ &= 2 + (n-1) \cdot 3. \end{aligned}$$

Recurrence relations: finding explicit formula

- Backtrack to find an explicit formula for the sequence defined by
  - the recurrence relation $b_n = 2b_{n-1} + 1$
  - initial condition: $b_1 = 7$
  - Sequence: ?

Solution

We begin by substituting the definition of the previous term in the defining formula.

$$\begin{aligned} b_n &= 2b_{n-1} + 1 \\ &= 2(2b_{n-2} + 1) + 1 \\ &= 2[2(2b_{n-3} + 1) + 1] + 1 \\ &= 2^3 b_{n-3} + 4 + 2 + 1 \\ &= 2^3 b_{n-3} + 2^2 + 2^1 + 1. \end{aligned}$$

A pattern is emerging with these rewritings of b_n. (*Note*: There are no set rules for how to rewrite these expressions and a certain amount of experimentation may be necessary.) The backtracking will end at

$$\begin{aligned} b_n &= 2^{n-1} b_{n-(n-1)} + 2^{n-2} + 2^{n-3} + \cdots + 2^2 + 2^1 + 1 \\ &= 2^{n-1} b_1 + 2^{n-1} - 1 && \text{using Exercise 3, Section 2.4} \\ &= 7 \cdot 2^{n-1} + 2^{n-1} - 1 && \text{using } b_1 = 7 \\ &= 8 \cdot 2^{n-1} - 1 \quad \text{or} \quad 2^{n+2} - 1. \end{aligned}$$

◆

Two useful summing rules were proved in Section 2.4. We record them again for use in this section.

S1. $1 + a + a^2 + a^3 + \cdots + a^{n-1} = \dfrac{a^n - 1}{a - 1}$

S2. $1 + 2 + 3 + \cdots + n = \dfrac{n(n+1)}{2}$

Recurrence relations in counting problems

- Someone deposits $10,000 in a savings account at a bank yielding 5% per year with interest compounded annually. How much money will be in the account after 30 years?

  - $A_n = 1.05A_{n-1}$
  - $A_n = (1.05)^n A_0$